Appendix C

Eigenvalues and Eigenvectors of a Tridiagonal Matrix

Consider a symmetric matrix of the form

$$A = \begin{pmatrix} a & b & & & & \\ b & a & b & & & \\ & b & a & b & & \\ & & & \ddots & & \\ & & & b & a & b \\ & & & & b & a \end{pmatrix}, \tag{C.1}$$

Theorem C.1 *Let $A \in \mathbb{R}^{(N-1)\times(N-1)}$ be as in* (C.1)*. Setting $h = 1/N$, the eigenvalues and eigenvectors of A are given by*

$$\mathbf{w}^{(k)} = \begin{pmatrix} w_1^{(k)} \\ \vdots \\ w_{N-1}^{(k)} \end{pmatrix}, \quad \text{where} \quad w_i^{(k)} = c \sin\left(k\pi i h\right), \qquad c \in \mathbb{R}, \qquad k = 1, \ldots N-1, \tag{C.2}$$

and

$$\lambda_k = a + 2b\cos(k\pi h), \qquad k = 1, \ldots N-1.$$

The eigenvectors are mutually orthogonal. If we set $c = \sqrt{2/(N-1)}$, the eigenvectors are orthonormal.

Proof In the following, we use the identity

$$\sin(\alpha \pm \beta) = \sin(a)\cos(\beta) \pm \cos(a)\sin(\beta) \tag{C.3}$$

A consequence of this identity is

$$\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2\sin(\alpha)\cos(\beta) \tag{C.4}$$

and

$$\sin(2\alpha) = 2\sin(\alpha)\cos(\alpha) \tag{C.5}$$

By definition of eigenvalues and eigenvectors, we need to show the following three statements:

1. $bw_{i-1}^{(k)} + aw_i^{(k)} + bw_{i+1}^{(k)} = \lambda_k w_i^{(k)} \qquad i = 2, \ldots, , N-2$

2. $aw_1^{(k)} + bw_2^{(k)} = \lambda_k w_1^{(k)}$
3. $aw_{N-1}^{(k)} + bw_{N-2}^{(k)} = \lambda_k w_{N-1}^{(k)}$

It is clear that eigenvectors are specified only up to a constant (i.e. if $\mathbf{w}$ is an eigenvector of A, so is $c\mathbf{w}$ for any (non-zero) $c \in \mathbb{R}$. Thus, without loss of generality, we may set $c = 1$ in (C.2)). We first show statement 1 by direct computation:

$$\begin{aligned} bw_{i-1}^{(k)} + aw_i^{(k)} + bw_{i+1}^{(k)} &= b\sin(k\pi(i-1)h) + a\sin(k\pi ih) + b\sin(k\pi(i+1)h) \\ &= 2b\sin(k\pi ih)\cos(k\pi h) + a\sin(k\pi ih) \\ &= \sin(k\pi ih)(2b\cos(k\pi h) + a) \\ &= w_i^{(k)}\lambda_k. \end{aligned}$$

Next we show 2:

$$\begin{aligned} aw_1^{(k)} + bw_2^{(k)} &= a\sin(k\pi h) + b\sin(k\pi 2h) \\ &= a\sin(k\pi h) + 2b\sin(k\pi h)\cos(k\pi h) \\ &= \sin(k\pi h)(2b\cos(k\pi h) + a) \\ &= w_1^{(k)}\lambda_k. \end{aligned} \tag{C.6}$$

In order to show statement 3, it is useful to first consider

$$\begin{aligned} w_{N-l}^{(k)} = \sin\left(k\pi(N-l)h\right) &= \sin\left(k\pi\frac{N-l}{N}\right) \\ &= \left(\cancelto{0}{\sin(k\pi)}\cos\left(\frac{k\pi l}{N}\right) - \cos\left(k\pi\right)\sin\left(\frac{k\pi l}{N}\right)\right) \\ &= (-1)^{k+1}\sin\left(\frac{k\pi l}{N}\right) \\ &= (-1)^{k+1}w_l^{(k)}, \end{aligned} \tag{C.7}$$

where we have used $\cos(k\pi) = (-1)^k$. Then we easily obtain

$$\begin{aligned} aw_{N-1}^{(k)} + bw_{N-2}^{(k)} = (-1)^{k+1}\left(aw_1^{(k)} + bw_2^{(k)}\right) &= (-1)^{k+1}w_1^{(k)}(2b\cos(k\pi h) + a) \\ &= (-1)^{2(k+1)}w_{N-1}^{(k)}(2b\cos(k\pi h) + a) \\ &= w_{N-1}^{(k)}\lambda_k. \end{aligned} \tag{C.8}$$

It remains to show that the eigenvectors can be taken to be orthonormal. Orthogonality is clear since the matrix is symmetric and the eigenvalues are distinct. We show by direct computation that with the normalization $c = \sqrt{2/(N-1)}$, the eigenvectors have unit magnitude. Using the identity $\sin^2(\alpha) = 1/2(1 - \cos(2\alpha))$ there holds

$$\begin{aligned} \left\|\mathbf{w}^{(k)}\right\|^2 &= \frac{2}{N-1}\sum_{i=1}^{N-1}\sin^2(k\pi ih) \\ &= \frac{1}{N-1}\sum_{i=1}^{N-1}(1 - \cos(2k\pi ih)) \\ &= 1 - \frac{1}{N-1}\sum_{i=1}^{N-1}\cos(2k\pi ih) \end{aligned}$$

Setting $q := exp(\hat{\imath}2\pi kh)$, where $\hat{\imath}$ is the imaginary unit, we can write

$$\sum_{i=1}^{N-1}\cos(2k\pi ih) = Re\left\{\sum_{i=1}^{N-1}e^{\hat{\imath}2\pi kih}\right\} = Re\left\{\sum_{i=1}^{N-1}q^i\right\}.$$

Using the standard formula for the geometric sum we obtain

$$\begin{aligned}
\left\|\mathbf{w}^{(k)}\right\|^2 &= 1 - \frac{1}{N-1} Re\left\{\sum_{i=1}^{N-1} q^i\right\} \\
&= 1 - \frac{1}{N-1} Re\left\{\frac{1-q^N}{1-q}\right\} \\
&= 1 - \frac{1}{N-1} Re\left\{\frac{1-e^{\hat{\imath}2\pi khN}}{1-e^{\hat{\imath}2\pi kh}}\right\}
\end{aligned}$$

But $hN = 1$, and thus $exp(\hat{\imath}2\pi khN) = \cos(k2\pi) + \hat{\imath}\sin(k2\pi) = 1$, which yields $\left\|\mathbf{w}^{(k)}\right\|^2 = 1$